Grade 10 · Algebra II · Topic 2: Quadratic Functions and Equations · Lesson 2-6 — The Quadratic Formula

Project Task — Jeddah Season Ticket Prices

A 20-minute group project. Work in fours. Every member signs the sheet.

Student name :	Date : .....
Group members :	

MATERIALS This sheet · Squared paper · A pencil and ruler · A calculator

THE DRIVING QUESTION

Two Jeddah Season venues sell tickets at different prices. Which venue can reach a revenue of 3 000 SAR, which can reach 4 000 SAR, and at what ticket prices?

THE SITUATION

A classroom model (invented for practice, not a real forecast): at Venue 1 the revenue is $R_1 = -10p^2 + 400p$ riyals and at Venue 2 it is $R_2 = -6p^2 + 300p$ riyals, where p is the ticket price in riyals.

The organisers need, for each venue and each target, the discriminant, the number of ticket prices that work and the prices themselves. Each group recommends a venue for each target.

$$R_1 = -10p^2 + 400p \qquad R_2 = -6p^2 + 300p$$

Venue	Target (SAR)	Discriminant	Prices (SAR)
1	3 000		
1	4 000		
2	3 000		
2	4 000		

YOUR TASK 5 steps · 20 minutes

1 REARRANGE (4 min)
For each row, set the revenue equal to the target and write a quadratic equal to zero. Write a, b, c with their signs.

2 DISCRIMINANT (4 min)
Find the discriminant for each row and say how many ticket prices work. Fill the table.

3 SOLVE (6 min)
Use the quadratic formula for every row with a positive or zero discriminant. Round to the nearest 0.01 SAR.

4 COMPARE (3 min)
Which venue can reach each target? Check one price by substituting into the model.

5 PRESENT (3 min)
One sentence for the organisers: how did the discriminant tell you the answer before you solved?

YOUR WORKING write on this sheet

Step 1 — the four equations:

Step 2 — the four discriminants:

Step 3 — the prices:

Steps 4 and 5 — our comparison and sentence:

ROLES
- Reader — states the models and the question
- Calculator — finds the discriminants
- Checker — substitutes prices into the model
- Presenter — says the sentence

DONE WHEN ...
- Every discriminant is written before any solving.
- A negative discriminant is explained as an impossible target.
- Units (SAR) appear in every answer.
- Your sentence makes sense to a non-mathematician.